

The empirical recurrence for OEIS A251447, recovered from its tail

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Abstract

OEIS A251447 records “Empirical recurrence of order 88” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 243$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 91$. Its last coefficient is $c_{88} = -108 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251447 is “Number of $(n+1) \times (4+1) 0..2$ arrays with no 2×2 subblock having the minimum of its diagonal elements less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

6505, 295339, 14434472, 711649307, 35114707166, 1733325153538, 85556587695052, 4223166671228639, ...

The entry states:

Empirical recurrence of order 88 (see link above).

As of the “Last modified” line on the live entry (Dec 26 2023 23:52:08, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 243$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 243$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 486$ exact terms from $n = 3$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	33	c_{23}	158576755066039459	c_{45}	−169669249418927397201	c_{67}	−24977050769
c_2	1071	c_{24}	658723325827721021	c_{46}	76574199911861309463	c_{68}	−49547353618
c_3	−8150	c_{25}	−1252163831740662278	c_{47}	24108707554065203781	c_{69}	1004702758
c_4	−260849	c_{26}	−2453820011134316109	c_{48}	−297963156662351367740	c_{70}	71316201302
c_5	554631	c_{27}	6541682056966825407	c_{49}	97506074716138286632	c_{71}	5339780278
c_6	25426015	c_{28}	5693780470125858230	c_{50}	357611006414848497097	c_{72}	−7891914326
c_7	−15030216	c_{29}	−23346689768663377385	c_{51}	−167298502601350386666	c_{73}	−1104748633
c_8	−1259532420	c_{30}	−5950126084534766196	c_{52}	−250682489699798340228	c_{74}	6859744215
c_9	837259989	c_{31}	57606202250608726577	c_{53}	151627176204571656049	c_{75}	1257848236
c_{10}	37161790539	c_{32}	−5898027594781113053	c_{54}	124505958942953291786	c_{76}	−542542110
c_{11}	−55149460798	c_{33}	−99252871385310960272	c_{55}	−93720713067549438730	c_{77}	−115652850
c_{12}	−675513827794	c_{34}	23860021625424763168	c_{56}	−55460366692231160658	c_{78}	382355886
c_{13}	1688934049527	c_{35}	127451882089067330117	c_{57}	46434191475223336998	c_{79}	81178280
c_{14}	5486761083656	c_{36}	−2526338344955417948	c_{58}	24342832698796498347	c_{80}	−19456096
c_{15}	−28666552405622	c_{37}	−157713249909586178075	c_{59}	−18651090229820550265	c_{81}	−2628934
c_{16}	33291865189860	c_{38}	−101143704053750214147	c_{60}	−10131211800906888968	c_{82}	7290303
c_{17}	255401680935115	c_{39}	238143492582245106885	c_{61}	5846619937515780599	c_{83}	−620178
c_{18}	−1337653607137677	c_{40}	232088612362359416101	c_{62}	3722285476251376923	c_{84}	−174304
c_{19}	−623853271501968	c_{41}	−329004721951773062996	c_{63}	−1393179331417223995	c_{85}	681203
c_{20}	16140906826499994	c_{42}	−265238271527835734171	c_{64}	−1124075957241000108	c_{86}	157013
c_{21}	−10913211693228317	c_{43}	306209998230739225919	c_{65}	238929673214978197	c_{87}	−5024
c_{22}	−121569894219414518	c_{44}	148828923276559795015	c_{66}	267829381552448851	c_{88}	−108

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 91$, and it is the recurrence OEIS A251447 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1x^{88-1} - \dots - c_{88}$. Its constant term is $-c_{88} = 108$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -108 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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